Lesson 2 Engel's thm and Lie's thm

Monday, March 2, 2026 17:11

Engel's thm

Def enveloping algebra

An associative algebra A is called an enveloping algebra of a Lie algebra L if $L \subseteq A^{(-)}$ and $A = \langle L \rangle$.

Thm

Let A be a finite dimensional enveloping algebra of L, i.e. $L \subseteq A^{(-)}$, $A = \langle L \rangle$. If $\exists m \geq 1$, s.t. $\forall a \in L$, $a^m = 0$, then A is nilpotent.

We say a subset $S \subset L$ is a Lie subset if $\forall a, b \in S$, $[a,b] \in S$ (not nec. a subspace)

In fact, we can prove a stronger version:

Let $S \subseteq L$ be a Lie subset of L, $L = \text{span } S$. If $\exists m \geq 1$, s.t. $\forall a \in S$, $a^m = 0$ then any f.d. enveloping algebra A is nilpotent.

note that if $\dim A = n < \infty$, A nilpotent $\Rightarrow A^{n+1} = (0)$ since $A \supsetneq A^2 \supsetneq \cdots \supsetneq A^{n+1} = (0)$ (at most codim 1)

pf:

By Zorn's lemma, we can choose a maximal Lie subset $S_1 \subset S$ s.t. the associative algebra $\langle S_1 \rangle$ is nilpotent $\Rightarrow \exists m$ s.t. $S_1{}^m = (0)$.

(for any chain $S_1 \subset S_2 \subset \cdots \subset$ in S s.t. $\langle S_i \rangle$ is nilpotent, take $S' = \bigcup S_i$, then S' is a Lie bracket ; $\langle S' \rangle$ is nilpotent since $\forall x_1, \dots, x_{n+1} \in \langle S' \rangle$, $\exists k$ s.t. $x_i \in \langle S_k \rangle$, thus $x_1 \cdots x_n = 0$. We conclude that $\langle S' \rangle^n = (0)$).

Then $(\operatorname{ad}(S_1))^{2n-1} = (0)$.

$(\operatorname{ad}(s_1) \cdots \operatorname{ad}(s_{2n-1})\, x = \sum_{p+q=2n-1} \pm s_{i_1} \cdots s_{i_p}\, x\, s_{j_1} \cdots s_{j_q})$

Claim: if $S_1 \neq S$, then $\exists s \in S \setminus S_1$ s.t. $[S_1, s] \subseteq S_1$

Pick any $s_1 \in S \setminus S_1$, if $\exists x_1 \in S_1$ s.t. $[x_1, s_1] \notin S_1$, let $s_2 = [x_1, s_1]$

If $\exists x_2 \in S_1$ s.t. $[x_2, s_2] \notin S_1$, let $s_3 = [x_2, s_2]$. $\cdots$

Since $(\operatorname{ad}(S_1))^{2n-1} = (0)$, the procedure ends in finite steps, i.e. s_k is the desired s for some k.

Now let $S_2 = S_1 \cup \{s\}$, a Lie subset of S

If we can show $\langle S_2 \rangle$ is nilpotent, then it contradicts with the maximality

Now let $S_2 = S_1 \cup \{s\}$, a Lie subset of S

If we can show $\langle S_2 \rangle$ is nilpotent, then it contradicts with the maximality of S_1, hence we conclude that $S_1 = S \Rightarrow A = \langle L \rangle = \langle S \rangle$ is nilpotent

To show $\langle S_2 \rangle$ is nilpotent, notice that any $0 \neq x \in \langle S_2 \rangle$, x is a linear combination of distinct nonzero products of the form

$$l = \underbrace{s_{11} s_{12} \cdots s_{1m_1}}_{\substack{s_{1i} \in S_1 \\ m_1 < n}} s^{k_1} \underbrace{s_{21} s_{22} \cdots s_{2m_2}}_{\substack{s_{2j} \in S_1 \\ m_2 < n}} s^{k_2} \underbrace{s_{31} \cdots}_{} \qquad \text{where } k_i < m.$$

Note that $s^{k_1} s_{21} = s^{k_1 - 1} s s_{21} = s^{k_1 - 1} [s, s_{21}] + s^{k_1 - 1} s_{21} s$

number of elements in S_1 remain unchanged.

Repeating, we can write l as a linear combination of products with s appearing in the end, hence $l \neq 0 \Rightarrow m_1 + m_2 + \cdots < n$

Thus such a product has length at most $n-1 + n(m-1)$

$$\_ S_1 \_ S_1 \_ \cdots \_ S_1 \_ \qquad n-1 \text{ elements in } S_1$$
$$n \cdot (m-1) \quad s$$

$\Rightarrow S_2^{nm} = (0)$.

Def module

Let L be a Lie algebra and M a vector space. Define an operation $L \times M \to M$. The following are equivalent:

$(a, m) \mapsto a \cdot m \in M$

1° $L + M$ is a Lie algebra if we define

$$\begin{cases} a \cdot m = -m \cdot a & \forall m \in M, a \in L \\ \text{multiplication in } M \text{ is trivial} \end{cases}$$

2° $[a, b] \cdot m = a(b \cdot m) - b(a \cdot m) \quad \forall a, b \in L, m \in M$

3° $L \to \mathrm{End}_F(M)^{(-)}$ is a homomorphism

We say M is a <u>module</u> over L if it satisfies any of these conditions.

Thm Engel's thm

Let L be a f.d. Lie algebra. Then

$\forall a \in L$, $\mathrm{ad}(a)$ is nilpotent $\Rightarrow L$ is nilpotent

pf: Note that since L is f.d., say of dimension $n < \infty$. Then

pf: Note that since L is f.d., say of dimension $n < \infty$. Then $\forall a \in L$, $\mathrm{ad}(a)^n = 0 \Rightarrow$ We can apply the above thm:
$\forall a \in L$, $(\mathrm{ad}\, L)^n = (0) \Rightarrow (\mathrm{ad}\, L)^n L = L^{n+1} = (0)$.

Rmk If L is infinite dimensional,

$\begin{cases} \text{ad-nilpotent of bounded } n \Rightarrow L \text{ nilpotent} \\ \text{not bounded} \not\Rightarrow L \text{ nilpotent} \end{cases}$

Rmk 3 different versions $(3) \Rightarrow (2) \Rightarrow (1)$

(1) $\dim_F L$, $\forall a \in L$, $\mathrm{ad}(a)$ is nilpotent $\Rightarrow L$ is nilpotent

(2) let M be a module over L, $\dim_F M < \infty$. (i.e. $L \xrightarrow{\varphi} \mathrm{End}_F M^{(-)}$)

$\forall a \in L$, $\underbrace{a(a(\cdots(a \cdot M)}_{n} = (0) \Rightarrow \exists N$, $\underbrace{L \cdots L}_{N} M = (0)$

(i.e. $\forall a \in L$, $\varphi(a)$ nilpotent $\Rightarrow \varphi(L)$ nilpotent)

$\Rightarrow \exists$ a basis in M: $L \subseteq \begin{pmatrix} 0 & & * \\ & \ddots & \\ 0 & & 0 \end{pmatrix}$

(3) $S \subseteq L$ Lie subset. $\forall a \in S$, $a^n M = (0) \Rightarrow \exists N$, $\underbrace{L \cdots L}_{N} M = (0)$.

Lie's thm

Lemma

Char $F = 0$ or char $F > n$

let a, b be two $n \times n$ matrices s.t. $[a,[a,b]] = 0$. Then $[a,b]$ is nilpotent

pf: Consider characteric polynomial of b:

$$b^n + \beta_{n-1} b^{n-1} + \cdots + \beta_1 b + \beta_0 = 0$$

Apply $\mathrm{ad}(a)^n$:

$\begin{cases} \mathrm{ad}(a)^n b^i = 0 & \forall i \le n-1 \text{ (decompose using } \mathrm{ad}\, a \in \mathrm{Der}\, L) \\ \mathrm{ad}(a)^n b^n = n!\,[a,b]^n & \text{(use induction)} \end{cases}$

$$\begin{aligned} (\mathrm{ad}\, a)^2 b^2 &= \mathrm{ad}\, a ([a, b^2]) \\ &= \mathrm{ad}\, a ([a,b]b + b[a,b]) \\ &= [a,[a,b]b] + [a, b[a,b]] \\ &= [a,[a,b]]b + [a,b]^2 + [a,b]^2 + b[a,[a,b]] \end{aligned}$$

$$= [a,[a,b]b] + [a, b[a,b]]$$

$$= [a,[a,b]]b + [a,b]^2 + [a,b]^2 + b[a,[a,b]]$$

$$= 2[a,b]^2$$

Thus we get $n!\,[a,b]^n = 0 \Rightarrow [a,b]^n = 0$.